MediVision AI: A Multimodal DenseNet121 Baseline for Multi-Label Chest Radiograph Classification, with an Audit of What Its Evaluation Record Can and Cannot Support

Y. Lakhani
Department of Computer Engineering, GMIU
Gujarat, India
yglakhani@gmiu.edu.in

***Abstract*—Deep-learning prototypes for chest radiograph interpretation are abundant, but the evidence they ship with is often thinner than their reported scores. This paper reports MediVision AI, a compact research baseline that couples a DenseNet121 image encoder to a small metadata encoder and classifies four findings on a patient-disjoint split of a 4,999-image subset of NIH ChestX-ray14. On 830 held-out studies, the recorded macro area under the ROC curve (AUC) is 0.771. Restricting the macro average to the three findings with at least 30 positive studies gives 0.791: Normal 0.724 (95% interval 0.690–0.758), Cardiomegaly 0.805 (0.744–0.867) and Pleural effusion 0.844 (0.798–0.891). Pneumonia has six positive studies and is reported descriptively only. Exact intervals for sensitivity and specificity are given throughout, and the small-event finding illustrates why they matter: its sensitivity is 0.667 with a 95% interval of 0.22–0.96. We then audit the evaluation record itself. Twenty stored metrics reproduce from the stored confusion counts to machine precision, but one archived ROC panel is inconsistent with those counts and is withdrawn rather than re-plotted. No image-only or metadata-only control was trained, so no fusion benefit is claimed; no threshold was selected on validation data; and per-study predictions were not retained, so patient-level bootstrap intervals remain unavailable. The contribution is a transparent baseline whose reported quantities are individually traceable to stored artifacts, together with an explicit account of the several things such a baseline cannot establish.**

***Index Terms*—**Chest radiography, class imbalance, DenseNet121, exact confidence intervals, label noise, multimodal fusion, reproducibility, shortcut learning.

I. INTRODUCTION

Chest radiography is the highest-volume diagnostic imaging examination worldwide, and the gap between examination volume and the availability of trained thoracic radiologists is widest in resource-constrained settings. Convolutional networks now reach radiologist-adjacent performance on selected chest radiograph tasks, which has produced a steady stream of prototype systems that convert a single frontal radiograph into finding-level probabilities. The empirical foundation of this literature is a small number of large public corpora annotated by mining radiology reports, of which NIH ChestX-ray14 and, more recently, MIMIC-CXR and CheXpert are the most frequently used [1]–[4].

The scientific value of such a prototype depends less on its headline area under the curve (AUC) than on whether its numbers can be traced to a defined model, a defined split and a defined evaluation script. Three failure modes recur. First, report-derived labels are noisy: the absence of a finding in a report is not evidence of its absence in the image, and multi-label targets built by string matching inherit this ambiguity [14]. Second, evaluation is frequently compromised either by patient-level leakage across splits or by silent changes of protocol between the training run and the reported table. Third, models readily exploit acquisition shortcuts—laterality markers, positioning devices, post-processing fingerprints and, in chest radiography, the anteroposterior versus posteroanterior projection—so that a measured association with disease may be an association with acquisition context [12], [13].

This paper reports MediVision AI, a four-finding chest radiograph baseline, and an audit of the evaluation record that accompanies it. We deliberately report the audit alongside the results rather than in a separate paper, because the two together determine what the numbers mean. The contributions are:

- **A fully specified, re-executable pipeline.** Data contract, patient-level splitting logic, transformation chain, normalisation constants, fusion architecture, objective and optimiser state are documented at the level of individual code paths (Section III), and every configuration value in the paper was read from the implementation rather than from earlier prose.
- **Held-out results with named denominators and exact intervals.** All metrics are reported on 830 patient-disjoint studies at a single threshold of 0.50, with per-finding counts that reproduce the reported sensitivities and specificities exactly. Clopper-Pearson intervals are reported

for every sensitivity and specificity and Hanley-McNeil intervals for every AUC (Sections IV-C, IV-D), which the earlier version of this work wrongly described as impossible; the Pneumonia interval, spanning 0.22–0.96, makes the small-event problem visible rather than rhetorical.

- **A restated macro average.** Because one modeled finding has six positive studies, we report the macro average over all four findings (0.771) together with a macro average restricted to the three findings that carry at least 30 positive studies (0.791), and we treat the second as the interpretable summary.
- **An audit of the evaluation record.** The stored metrics are arithmetically closed, but one archived ROC panel cannot describe the predictions summarized by the counts, and it is withdrawn from the paper rather than redrawn. Section IV-G states the test that any future figure must pass.

The scope statement matters as much as the results. MediVision AI is a research prototype. Its outputs are model probabilities, not diagnoses; its Grad-CAM maps are attention diagnostics, not localizations of disease; and the framework has had no reader study, no external validation and no calibration analysis. The multimodal design is described as a design, not as a demonstrated benefit: no image-only control was trained, so the paper makes no claim that metadata improves performance.

II. RELATED WORK

A. Public Corpora and Report-Derived Labels

ChestX-ray14 introduced 112,120 frontal radiographs from 30,805 patients with fourteen finding labels extracted from reports by a rule-based labeller [1]; the subset used here is drawn from that corpus. MIMIC-CXR provides 377,110 images from 227,835 studies with paired free-text reports [2], and CheXpert contributes 224,316 radiographs with an explicit uncertainty category and a radiologist-annotated validation and test set [3]. The choice between corpora is essentially a choice of label-noise model: report-mined labels overstate prevalence, understate rare findings, and introduce systematic differences between the annotation process and the clinical question being studied; the concrete failure modes of the ChestX-ray14 labeller have been analysed in detail [10]. CheXNeXt showed that a carefully validated model can reach radiologist-level performance on selected findings [4], but that result rests on a purpose-built radiologist-annotated test set, which is precisely the resource most prototypes lack.

B. Architectures for Chest Radiograph Classification

DenseNet121 has become a common backbone for chest radiograph classification because dense connectivity reuses features aggressively in the data regime typical of medical imaging [5]; residual networks remain the standard alternative [6], and ImageNet pre-training is the default initialization for both [7]. At the scale studied here the classifier head, the input resolution and the fusion strategy plausibly dominate the reported performance more than the choice between densely and residually connected backbones, although this study does not test that claim: it uses one backbone and one resolution. Transfer learning from natural-image pre-training is the dominant recipe across medical imaging modalities, and the small-data regime in which it is used has been documented for retinal imaging as well as for thoracic imaging [22].

C. Label Noise and Weak Supervision

Karimi et al. survey noise-robust training for medical image analysis and distinguish noise in feature space from noise in label space, observing that symmetric noise can be partly absorbed by regularization whereas asymmetric, class-dependent noise—the regime created by report mining—biases decision boundaries systematically [14]. For chest radiography the dominant asymmetry is that positive mentions are comparatively reliable while negative evidence is inferred from silence. Framing the task as multi-label sigmoid classification, as done here, avoids forcing a single mutually exclusive class but does not remove the underlying ambiguity.

D. Shortcut Learning and Spurious Correlation

Geirhos et al. define shortcut learning as the tendency of deep networks to adopt decision rules that perform well on a benchmark while violating the intended solution [12]. In chest radiography the canonical example is Zech et al., who showed that a pneumonia classifier transferred across institutions lost most of its apparent performance, and attributed the drop to hospital-specific markers and acquisition differences rather than to disease morphology [13]. Two consequences follow. First, a metadata channel that encodes view position, image dimensions and pixel spacing is a plausible shortcut carrier. Second, internal validation on a single-institution corpus cannot distinguish a learned disease signature from a learned acquisition signature, which is why external validation is listed as a hard limitation in Section V.

E. Calibration and Subgroup Fairness

Discrimination and calibration are distinct properties, and a model can rank cases well while producing probabilities that are systematically over- or under-confident; Van Calster et al. describe calibration as the weak point of clinical prediction models and recommend explicit calibration curves rather than AUC-only reporting [15]. Equity is the second axis: Seyyed-Kalantari et al. report underdiagnosis bias of chest radiograph classifiers in under-served populations [16], Larrazabal et al. show that gender imbalance in training data

produces biased classifiers [17], Obermeyer et al. document how a proxy target can encode a protected attribute even when that attribute is never a feature [18], and Gichoya et al. demonstrate that patient race is recoverable from medical images alone [19]. A framework that ingests patient age and sex, as this one does, cannot claim subgroup neutrality by construction, and no subgroup analysis is reported here because the evaluation record does not retain per-patient predictions.

F. Multimodal Fusion of Imaging and Clinical Data

Baltrusaitis et al. provide the standard taxonomy of multimodal learning, distinguishing early, late and jointly learned fusion [20]. Huang et al. review fusion of medical imaging with electronic health records and emphasize that the incremental value of clinical variables is frequently confounded with acquisition metadata that co-varies with the clinical variable [21]. MediVision AI implements a late-fusion design in which a 32-dimensional metadata embedding is concatenated with the pooled image descriptor. This is the cheapest possible fusion and, as Section IV-F argues, the design most exposed to shortcut acquisition of the metadata channel. Whether it helps at all is untested here.

G. Explainability and Its Limits

Grad-CAM produces coarse localization maps from the gradient of a target score with respect to a convolutional feature map [8], building on class activation mapping [9]. Such maps are widely presented as evidence of clinical plausibility, which overstates what a saliency map can establish: Rudin argues that post-hoc explanations of black-box models are unreliable substitutes for interpretable models in high-stakes decisions [11], and saliency maps are in general not unique, not stable under small perturbations and not formally faithful to the decision function. This framework therefore exposes Grad-CAM as a qualitative inspection aid for developers and not as a validation instrument. No saliency map is shown in this paper, because no radiograph and no checkpoint are part of the release; see Section V.

III. MATERIALS AND METHODS

The pipeline of Fig. 1 is summarized below. It covers the framework. The subsections below cover data assembly and partitioning (A–C), the multimodal encoder and its objective (D–E), and optimization and evaluation (F–H).

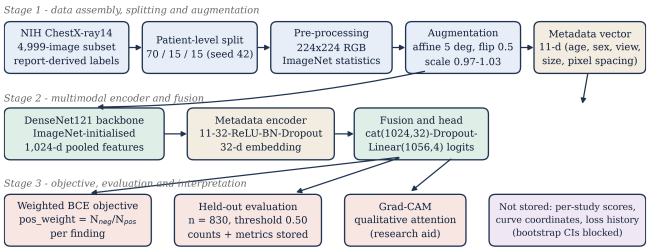


Fig. 1. Framework as implemented. Stage 1 assembles the metadata table, partitions it by patient and builds the augmentation chain; Stage 2 encodes images and metadata separately and fuses the descriptors; Stage 3 applies the weighted objective, evaluates at a fixed threshold and produces qualitative explanations. The dashed box marks the control experiments the release does not contain.

A. Data Source and Label Construction

The study uses a 4,999-image working subset of the NIH ChestX-ray14 corpus [1], retained in the original NIH comma-separated metadata format with the columns *Image Index*, *Finding Labels*, *Patient ID*, *Patient Age*, *Patient Sex*, *View Position*, image dimensions and pixel spacing. Four findings are modeled: *Normal*, *Pneumonia*, *Cardiomegaly* and *Pleural effusion*. Binary targets are derived from the pipe-delimited *Finding Labels* field by a deterministic alias map, in which the literal token *No Finding* maps to Normal and the tokens *Effusion*, *Pleural Effusion* and *Pleural_Effusion* map to Pleural effusion. Because the task is multi-label, a study may be positive for more than one modeled finding, and the Normal indicator is treated as an ordinary target rather than as the complement of disease. Patient identifiers, study identifiers and follow-up indices are retained for traceability and excluded from all model inputs.

Two properties of this construction bound what the Normal label means. It is not an independent radiological judgment: a study is Normal when the derived label vector is positive for *No Finding*, which is compatible with the report mentioning findings outside the four modeled here. And the modeled set is a four-of-fourteen projection of the corpus, chosen for tractability rather than by a principled selection rule; the resulting prevalences are reported in Table I and are specific to this subset. The released artifacts do not record how the 4,999-image subset was drawn (whether by random sampling, by stratification, or by taking a published Kaggle split), so the subset cannot be shown to be representative of ChestX-ray14, and the paper makes no such claim.

TABLE I
Corpus and held-out composition. Split sizes are images; the nominal 70/15/15 allocation is applied to patients, and the realized image fractions differ because studies per patient vary.

Partition / finding	Images	Share	Positives	Prevalence
Training	3,447	68.95%	—	—
Validation	722	14.44%	—	—
Test	830	16.60%	643[a]	—
Normal	830	—	451	0.543
Pneumonia	830	—	6	0.007
Cardiomegaly	830	—	73	0.088
Pleural effusion	830	—	113	0.136
Total	4,999	100%	—	—

Note: [a] Positive label decisions summed across the four findings; the targets are not mutually exclusive, so this exceeds the number of studies. Per-finding positive counts for the training and validation partitions are not retained in the released metric record and are therefore not reported; the training prevalences that determine the weights of Equation (3) cannot be quoted for the same reason.

B. Patient-Level Partitioning

Partitioning is performed on the unique patient identifier, not on the image. The patient list is shuffled with a fixed seed and split 70%, 15% and 15%; the image table is then filtered by membership. Splitting at the image level would let repeat examinations of the same patient appear on both sides of the split and would inflate held-out performance, an error that is common in chest radiograph benchmarks because the corpus contains several images per patient [1]. The realized partitions are 3,447 training, 722 validation and 830 held-out images (Table I). The number of distinct patients is not stored in the released record, so the effective sample size of every interval in this paper is bounded below by the study count and above by the patient count; intervals are computed on studies and therefore ignore within-patient clustering, which makes them narrower than a patient-level interval would be. This is stated as a caveat wherever an interval is quoted.

C. Transformations and Augmentation

Training inputs are resized to 224 × 224, augmented with a random affine transform (rotation ±5°, translation 2%, scale 0.97–1.03) and a random horizontal flip with probability 0.5; validation and test inputs receive resize and tensor conversion only. Both branches then apply channel-wise standardization with the ImageNet statistics $\mu = (0.485, 0.456, 0.406)$ and $\sigma = (0.229, 0.224, 0.225)$ [7]. Two properties deserve comment. First, the chain contains no intensity, contrast, gamma or histogram-equalization step, so the pipeline does not attempt to normalize the acquisition-dependent contrast differences that motivate external-validation failures [13]. Second, horizontal flipping is not label-preserving for every thoracic finding: it exchanges left and right, and Cardiomegaly in particular is a laterality-sensitive judgment because the cardiac silhouette is positioned on one side. Flipping was applied to all findings in training, and its effect on this finding is not measured here; it is recorded as a design risk in Section IV-F rather than as a verified harm.

D. Network Architecture

The image branch is DenseNet121 pre-trained on ImageNet, truncated at the global average pooling layer, giving a 1,024-dimensional descriptor. The metadata branch encodes an 11-dimensional vector comprising age, original image width and height, x and y pixel spacing, one-hot encoding of patient sex (two levels plus an unknown bucket) and one-hot encoding of view position (posteroanterior, anteroposterior, unknown). Numeric metadata are scaled by fixed constants (age by 100, dimensions by 3,000, spacing by unity) so that each component lies approximately in [0, 1]. A two-layer encoder maps the metadata vector to a 32-dimensional embedding; the two descriptors are concatenated and passed to a linear classifier with dropout. Writing f_θ for the image encoder, g_φ for the metadata encoder and W, b for the head, the model computes

$$z = W\,[\,\mathrm{avgpool}(f_\theta(x))\,;\,g_\varphi(m)\,] + b, \qquad (1)$$

$$p_c = \sigma(z_c)$$

where x is the image tensor, m the metadata vector and σ the element-wise logistic function. When metadata are unavailable at inference time the branch receives a zero vector, which yields the encoder's learned bias response rather than an error; the service therefore degrades gracefully, but a missing-metadata prediction is not distributionally identical to a complete one.

E. Objective and Class Imbalance

The objective is a class-weighted binary cross-entropy over the four findings. For finding c, let $N_{pos,c}$ and $N_{neg,c}$ be the numbers of positive and negative training images and let w_c be the corresponding positive-class weight. The training loss is

$$L = -(1/N) \sum_i \sum_c [\, w_c\, y_{ic} \ln p_{ic} + (1 - y_{ic}) \ln(1 - p_{ic}) \,] \qquad (2)$$

$$w_c = N_{neg,c} / N_{pos,c} \qquad (3)$$

with $p_{ic} = \sigma(z_{ic})$ and $y_{ic} \in \{0, 1\}$. The weighting is computed from the training partition only, and denominators are floored at one to avoid division by zero. Three consequences follow from Equation (3) directly.

- **Positive weighting does not change ranking metrics.** Because $w_c > 1$ is a strictly increasing rescaling of the positive term, it moves the iso-contour at a fixed threshold but not the ordering of scores within a finding. ROC-AUC is therefore invariant to the weighting, and any improvement in AUC attributed to w_c could only

be an artifact of training dynamics—different gradient magnitudes, hence a different parameter trajectory—not of the metric.

- **The weighting does move the operating point, and not necessarily in a useful direction.** The weight is largest where a finding is rarest, so the fixed 0.50 threshold lands at very different positions on different ROC curves. The largest weight belongs to the rarest modeled finding, which is why the operating point of the rarest finding is the least interpretable in the results; the training-partition positive counts that determine the individual weights are not retained in the released record, so the ordering of the weights can be inferred from the test prevalences only approximately and is not reported as a measured quantity.
- **The weighted loss is not comparable across findings.** Validation loss is a weighted quantity and must not be compared across findings or against unweighted literature values. It is used only for checkpoint selection within a single run.

F. Optimization and Training Configuration

Training uses AdamW [23] with decoupled weight decay, a base learning rate of 1×10^{-4}, weight decay 1×10^{-4}, mini-batches of 16 and a ReduceLROnPlateau schedule (factor 0.5, patience 2). Early stopping is configured with patience five epochs, and the checkpoint rule keeps the parameter set with the lowest weighted validation loss. Twenty epochs were requested and eight were completed; the stop is recorded as a completed run of eight epochs, not as a convergence decision, because the stopping criterion was never satisfied and no terminal event is logged. The model is therefore best read as partially trained: eight passes of a 3,447-image partition with one frozen-backbone epoch is a small budget, and the held-out AUCs reported here should be treated as an early-run result rather than a converged one. A single seed (42) controls patient splitting, weight initialization and the augmentation stream, and no replication across seeds is reported. Table II summarizes the configuration as read from the code.

TABLE II
Training configuration as read from the implementation.

Parameter	Value	Source of record
Optimizer	AdamW (β_1 0.9, β_2 0.999)	code
Learning rate	1×10^{-4}	code / config
Weight decay	1×10^{-4}	code / config
Batch size	16	code / config
Epochs requested / completed	20 / 8	config / run record
Early-stopping patience	5 epochs (weighted validation loss)	code / config
LR schedule	ReduceLROnPlateau (factor 0.5, patience 2)	code
Random seed	42 (split, init, augmentation)	code / config
Patient-level split	70 / 15 / 15	code
Input resolution	224 × 224 × 3	code / metadata
Normalization	ImageNet mean and SD	code
Augmentation	Affine ±5°, translate 2%, scale 0.97–1.03; h-flip p = 0.5	code
Dropout	0.20 (metadata encoder, head)	code
Loss	BCEWithLogits, pos_weight = N_{neg}/N_{pos}	code
Threshold	0.50 (fixed, not tuned on test)	code

Note: The completed-epoch count is eight of a requested twenty. The run stopped short of its budget with the stopping criterion unsatisfied and no terminal event logged, so the model is described as partially trained rather than converged.

One design decision in Table II deserves explicit criticism. Checkpoint selection minimizes the weighted validation loss of Equation (2). Because that loss sums four weighted per-finding terms, the finding with the largest weight dominates the selection criterion, and the validation partition has few positives for the rarest findings. A macro-averaged validation AUC would be a better selection criterion for a ranking problem, and its absence is a design weakness of the reported run rather than a property of the data.

G. Evaluation Metrics and Intervals

For each finding, with TP, TN, FP and FN denoting the entries of the 2 × 2 count matrix at threshold 0.50,

$$\text{Sens} = \text{TP}/(\text{TP}+\text{FN}), \quad \text{Spec} = \text{TN}/(\text{TN}+\text{FP}), \tag{4}$$

$$\text{Prec} = \text{TP}/(\text{TP}+\text{FP}), \quad \text{Acc} = (\text{TP}+\text{TN})/\text{N}$$

$$\text{F1} = 2 \cdot \text{Prec} \cdot \text{Sens}/(\text{Prec}+\text{Sens}) \tag{5}$$

and ROC-AUC is computed by the rank-based estimator. Macro aggregates are unweighted means over findings; micro aggregates pool the four 2 × 2 matrices into one count matrix, so a study contributes four label decisions. Macro and micro values therefore answer different questions—equal weight per finding versus equal weight per label decision—and both are reported because the corpus

is severely imbalanced across findings (Section IV-A).

Two kinds of interval are reported, and neither requires per-study scores. For a sensitivity k/n the exact Clopper-Pearson interval is obtained by inverting the binomial test, using the beta quantiles $B_{\alpha/2}(k, n-k+1)$ and $B_{1-\alpha/2}(k+1, n-k)$; Wilson score intervals are reported alongside in the supplementary record and differ from the exact intervals by less than 0.01 for all cells except Pneumonia. For an AUC with n_{pos} positives and n_{neg} negatives, the Hanley-McNeil variance approximation is

$$SE^2(A) = [\, A(1-A) + (n_{pos}-1)(Q_1-A^2) + (n_{neg}-1)(Q_2-A^2) \,] / (n_{pos}n_{neg}), \quad (6)$$

with $Q_1 = A/(2-A)$ and $Q_2 = 2A^2/(1+A)$. Both constructions assume independent observations, so they ignore within-patient clustering and are slightly optimistic for that reason; the assumption is stated rather than corrected, because correcting it requires the patient identifiers that the record does not retain. Neither construction supports a comparison between two models, which would require paired resampling. Every interval quoted in Section IV is computed by the routine described here, and no interval is quoted anywhere in this paper that is not.

A third quantity is reported for the AUCs and is not an interval in the statistical sense. The count matrix of each finding pins its ROC curve to a single point (FPR, TPR). Any non-decreasing curve from (0, 0) to (1, 1) through that point has an area bounded below by the piecewise-linear path through it and above by the path that rises to TPR = 1 before the point and stays there:

$$A_{min} = \½f\cdot t + (1-f)t, \;\; A_{max} = f\cdot t + (1-f), \quad (7)$$

where f and t are the verified false- and true-positive rates. Both bounds are attainable, so Equation (7) gives the exact range of AUCs compatible with the counts, and a recorded AUC outside that range is not merely unlikely but impossible. This is the test applied in Section IV-G; because the bounds depend on a single operating point they are wide, and they are not a substitute for an interval.

H. Implementation

The reference implementation is written in Python with PyTorch [24] and torchvision for the backbone and transformation chain, scikit-learn for the rank statistics and count matrices, and matplotlib for figure generation. Training checkpoints carry the model state, optimizer and scheduler state, epoch index, validation loss and the running per-epoch history, which allows an interrupted run to resume from the last completed epoch. Evaluation is a separate entry point that reloads a saved checkpoint, reconstructs the deterministic test partition from the same seed and writes a structured JSON record. The separation is deliberate: the evaluation script never fits, tunes or selects anything, so the reported numbers cannot be contaminated by test-time decisions.

IV. RESULTS

A. Corpus and Partition Composition

Table I gives the realized partition sizes and the held-out prevalence of each modeled finding. Two features dominate the interpretation of every later table. First, the findings are not merely imbalanced but separated by nearly two orders of magnitude: 451 of 830 held-out studies are labeled Normal whereas only 6 are labeled Pneumonia. Second, the positive labels are not mutually exclusive, so the 830 studies contribute 643 positive label decisions across the four findings. Any unweighted macro average therefore contains a term estimated from six events, and its sampling variability is dominated by that term.

B. Training Behaviour

The released record states the number of completed epochs, eight, together with the evaluated metrics. It states nothing about the shape of the loss trajectory: the training loop accumulates a per-epoch history in memory and writes it only into a checkpoint, and no checkpoint is part of the release. No training-curve figure can therefore be drawn from the released artifacts, and this paper contains none. What is claimed is limited to two facts recorded in the run: eight of the twenty budgeted epochs completed, and the early-stopping criterion was never satisfied during those eight epochs. The practical implication is stated in Section III-F: the model is partially trained, and its held-out numbers should be read as an early-run result.

C. Held-Out Discrimination

Table III reports per-finding performance on the 830 held-out studies at the fixed threshold of 0.50, and Fig. 2 shows the operating points and the AUC intervals. Discrimination is moderate and heterogeneous. Pleural effusion is the best-ranked finding (AUC 0.844, 95% interval 0.798–0.891) and Normal the weakest of the three substantial findings (0.724, 0.690–0.758). Cardiomegaly sits between them at 0.805 (0.744–0.867), and the two Cardiomegaly and effusion estimates are close enough that their intervals nearly touch, so the paper does not rank them. Pneumonia's recorded AUC of 0.709 rests on six positive studies; its interval (0.474–0.944) spans almost the entire admissible range, and it is reported for completeness only.

TABLE III
Held-out performance at a fixed threshold of 0.50 (n = 830 studies). AUC is threshold-free; the other columns are computed at the fixed point. The 95% intervals for every AUC, sensitivity and specificity appear in Table IV.

Finding	AUC	Sens.	Spec.	Prec.	F1
Normal	0.724	0.530	0.792	0.752	0.622
Pneumonia[b]	0.709	0.667	0.518	0.010	0.020
Cardiomegaly[c]	0.805	0.479	0.863	0.252	0.330
Pleural effusion	0.844	0.876	0.646	0.280	0.425
Macro (4)	**0.771**	**0.638**	**0.705**	**0.323**	—
Macro (3)[d]	**0.791**	**0.628**	**0.767**	**0.428**	—
Micro	—	0.586	0.688	0.311	0.407

Note: [b] Six positive studies. Every quantity in this row is descriptive; the sensitivity interval spans 0.22–0.96 and a single decision moves it by 0.167. [c] The AUC is the value in the archived metric record; its ROC panel was not archived in a form consistent with the counts and is withdrawn (Section IV-G). [d] Unweighted mean over Normal, Cardiomegaly and Pleural effusion, the findings with at least 30 positive studies. Micro aggregates pool the four count matrices. No interval is reported for a macro average: the variance of an average of dependent per-finding estimates cannot be computed from aggregate counts, and no pairing information survives in the record.

TABLE IV
95% intervals for the held-out estimates of Table III. Sensitivity and specificity intervals are exact (Clopper-Pearson); AUC intervals use the Hanley-McNeil variance of Eq. (6). All four constructions assume independent studies and therefore ignore within-patient clustering.

Finding	AUC 95% CI	Sens. 95% CI	Spec. 95% CI
Normal	0.690–0.758	0.483–0.577	0.747–0.831
Pneumonia[b]	0.474–0.944	0.22–0.96	0.483–0.553
Cardiomegaly	0.744–0.867	0.361–0.600	0.836–0.886
Pleural effusion	0.798–0.891	0.801–0.931	0.609–0.681

Note: The Pneumonia row is the reason intervals are reported at all: its sensitivity interval spans 0.22–0.96 on six positive studies, so no statement about its operating point can be supported. Intervals for the macro averages are not reported; see the note to Table III.

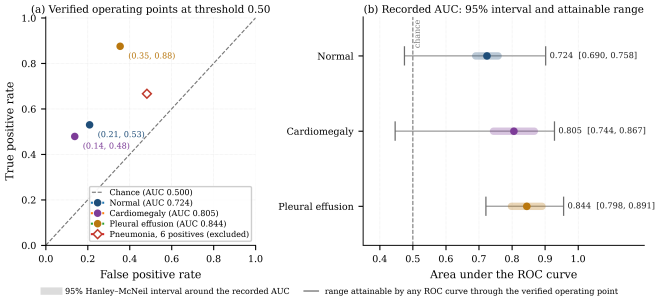


Fig. 2. Discrimination evidence for the held-out partition (n = 830). (a) Verified operating points at threshold 0.50, i.e. the single (FPR, TPR) pair implied by each count matrix. No curve is drawn: aggregate counts do not determine the interior of a ROC curve, and no curve coordinates were archived. (b) Recorded AUC per finding (dot), its 95% Hanley-McNeil interval (bar) and the exact range of AUCs attainable by any monotone curve through the verified operating point (grey bracket, Eq. 7). The recorded value lies inside its attainable range for all three findings; the Pneumonia AUC is omitted because its six positive studies make the interval uninformative.

The macro AUC over all four findings, 0.771, is the arithmetic mean of the four recorded values and inherits the six-event Pneumonia term. Restricting the mean to the three findings with at least 30 positive studies gives 0.791 with a macro sensitivity of 0.628 and a macro specificity of 0.767 at the fixed threshold; the corresponding all-finding values are 0.638 and 0.705. We report both and treat the three-finding average as the interpretable summary, because an unweighted mean that gives a sixth of its weight to a finding with six positives is not a summary of anything stable. Neither value is comparable to AUCs reported on radiologist-annotated test sets [4] or on the full fourteen-finding benchmark [1]: the cohort, the label construction and the finding set all differ.

D. Operating-Point Behaviour and Intervals

Table V decomposes the same results into confusion counts, and Fig. 3 shows the four matrices. The macro sensitivity of 0.638 against a macro specificity of 0.705 describes a model that is conservative on Cardiomegaly (sensitivity 0.479 at specificity 0.863) and permissive on Pleural effusion (sensitivity 0.876 at specificity 0.646). The operating points are the best-corroborated quantities in the study, because they follow arithmetically from count matrices that each account for all 830 studies.

TABLE V
Verified confusion counts at threshold 0.50, with the operating point they imply and the range of AUCs attainable by any monotone ROC curve through that point (Eq. 7). Every matrix accounts for all 830 held-out studies; the operating point is (FPR, TPR) = (FP/(FP+TN), TP/(TP+FN)).

Finding	TN	FP	FN	TP	AUC range
Normal	300	79	212	239	0.475–0.902
Pneumonia	427	397	2	4	0.506–0.839
Cardiomegaly	653	104	38	35	0.447–0.928
Pleural Effusion	463	254	14	99	0.721–0.956
Pooled	**1843**	**834**	**266**	**377**	—

Note: Counts and every rate derived from them were recomputed from the released record and agree exactly. The attainable range is the exact interval of ROC areas compatible with a single operating point; it is wide by construction and is a consistency check, not a confidence interval. Normal's operating point is (0.208, 0.530), Pneumonia's (0.482, 0.667), Cardiomegaly's (0.137, 0.479) and Pleural effusion's (0.354, 0.876). The recorded AUC of every finding lies inside its own range.

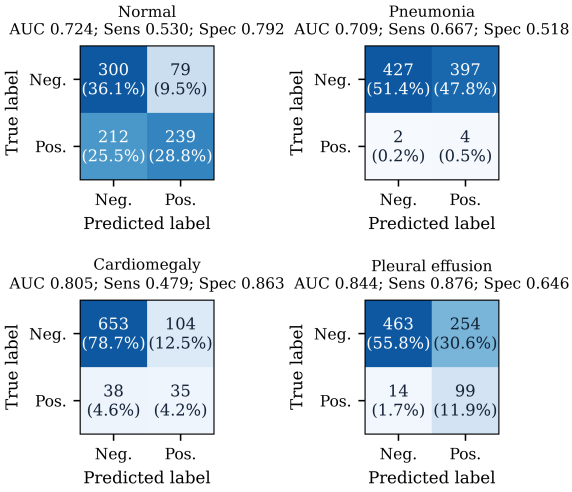


Fig. 3. Confusion matrices at a fixed threshold of 0.50, one 2 × 2 matrix per finding, with row-normalized percentages. Counts match Table V exactly. The Pneumonia matrix should be read with the six-event denominator in mind.

With intervals attached, three of these operating points sharpen and one collapses. Pleural effusion sensitivity 0.876 (0.801–0.931) and specificity 0.646 (0.609–0.681) are estimated precisely enough to support the statement that the model runs at a high-sensitivity, low-specificity point for this finding. Cardiomegaly sensitivity 0.479 (0.361–0.600) does not support a claim of conservative behaviour in either direction with confidence, although its specificity 0.863 (0.836–0.886) is well estimated. Pneumonia is uninterpretable at this denominator: sensitivity 0.667 with a 95% interval of 0.22–0.96, and specificity 0.518 (0.483–0.553) estimated from a class in which the model predicts positive for roughly half of all negatives. A single decision moves Pneumonia sensitivity by 0.167, and precision of 0.010 reflects four correct positives among 401 positive predictions.

The Normal finding produces 212 false negatives: studies labeled Normal that the model flags as abnormal. Because the four targets are independent sigmoids, this is the signature of a model that over-predicts disease when the label set is dominated by negative evidence, and it is the largest single contributor to the gap between per-finding accuracy (0.649) and the specificity of the disease findings. Pleural effusion generates 254 false positives, the price of a high positive weight. Precision is low for every finding except Normal, which follows directly from low prevalence and is the reason precision and recall must be read together at a stated prevalence.

E. What the Counts Imply About the Curves

Because the pipeline stores only aggregate metrics, the interior of each ROC curve is not recoverable from the release: no per-study score vector, and no false- or true-positive-rate array, survives in it. Aggregated metrics do not determine the shape of a ROC curve, and this paper therefore does not plot one. What the counts do determine is (i) the single operating point of each finding, (ii) the exact range of AUCs compatible with that point, given by Equation (7), and (iii) intervals from the binomial and Hanley-McNeil constructions of Section III-G. The first two are shown in Fig. 2, and all three are tabulated in Table IV and Table V.

The attainable range in Fig. 2(b) is reported because it converts a qualitative doubt into a checkable inequality. For Normal the recorded AUC of 0.724 lies inside the range [0.475, 0.902]; for Cardiomegaly 0.805 lies inside [0.447, 0.928]; and for Pleural effusion 0.844 lies inside [0.721, 0.956]. A recorded value that fell outside its own range would be arithmetically impossible rather than merely imprecise. This test, not a visual comparison of curves, is what the audit in Section IV-G rests on.

F. Design Risks and What This Study Cannot Say

The most consequential design risk in this framework is that the metadata encoder receives view position as a one-hot input. Posteroanterior and anteroposterior projections are not exchangeable: anteroposterior views are acquired predominantly in patients who are bed-bound, portable or acutely unwell, whereas posteroanterior views are acquired in ambulatory patients standing at a dedicated unit. The projection is therefore correlated with illness severity, examination setting and the reporting pathway that produces the label. A model that learns ‘anteroposterior implies disease’ can reduce loss without learning anything about lung morphology, and the correlation is strong enough that the behaviour would not be visible in an aggregate AUC. Image dimensions and pixel spacing carry a weaker but similar signal, since they fingerprint the acquisition device.

This risk has two implications the present study cannot resolve. First, any performance gain attributed to metadata fusion must be read as an acquisition-context effect until the opposite is demonstrated, and the demonstration requires an ablation comparing image-only, metadata-only and fused models on the same partition with paired intervals. No such ablation exists here: a single fused model was trained, and no control checkpoints were registered. The framework therefore supports no claim about fusion benefit in either direction, and the paper makes none. Second, a metadata-only control is the cheapest available diagnostic—if a model that never sees the image attains non-trivial AUC, the fused model's gain is contaminated by construction—and running it is listed as outstanding work in Section V. The permutation audit described below is likewise specified and not run: score each held-out study twice, once with its observed metadata and once with the view-position field permuted across studies, and report the change in AUC per finding.

Three further design elements are recorded here as risks rather than results. Horizontal flipping is applied to Cardiomegaly despite the finding's laterality dependence, and the effect is unmeasured. The fixed threshold of 0.50 is not a

decision rule: the validation partition exists, was used for checkpoint selection, and was not used to choose per-finding thresholds or to produce precision-recall curves, which are the appropriate summary at these prevalences. And the Normal label is not an independent radiological category, as Section III-A explains, so Normal performance partly measures the model's ability to recognize findings it does not model.

G. Provenance of the Archived ROC Figure

The release contains a ROC figure and a confusion figure alongside the JSON metric record. The metric record is arithmetically closed: for each finding, the stored count matrix reproduces the stored sensitivity, specificity, precision and F1, every matrix sums to 830, and the stored macro AUC equals the mean of the four per-finding AUCs to six decimal places. The confusion figure agrees with those matrices in the rank order of its cell shadings for all four findings. The ROC figure is the exception. It stores no numeric coordinates—the pipeline writes only a rendered image—so it can be measured but not recomputed, and for Cardiomegaly what it yields is incompatible with the rest of the record: the archived curve reaches a true-positive rate of about 0.96 at the false-positive rate of 0.137 where the counts place the operating point at 0.479.

Equation (7) settles the question. A monotone ROC curve through (0.137, 0.479) has an area between 0.447 and 0.928. The recorded AUC of 0.805 lies in that range; the value obtained by integrating the archived curve for this finding, 0.957, does not. The archived panel is therefore withdrawn from this paper rather than redrawn. The reported Cardiomegaly AUC remains 0.805, the value in the internally consistent record.

Two statements bound what this resolution establishes. First, it does not re-derive the reported value: no curve was stored for Cardiomegaly, and re-establishing one requires re-running evaluation, which the release does not permit because the checkpoint and the image corpus are excluded from version control. The value is reported as recorded and arithmetically consistent, not as independently recomputed. Second, the fault is confined to the archived panel: the counts, the metric record and the confusion figure agree with each other, so the discrepancy is most consistent with a stale or partially regenerated panel rather than with a corrupted evaluation. The test that any replacement figure must pass is stated here: its operating point must reproduce the counts to within 0.01, and its integral must equal the recorded AUC to within 0.01.

H. Statistical Analysis: What Is and Is Not Reported

The protocol called for 95% confidence intervals from patient-level bootstrap resampling of the held-out studies over 2,000 replicates, and for a paired bootstrap over 2,000 replicates to quantify the difference between the fused model and an image-only control. Neither is available, for a checkable reason: the evaluation entry point writes per-finding aggregate metrics and a count matrix but discards the per-study predicted probabilities, so the resampling unit—the study, with its patient identifier and label vector—cannot be reconstructed from the release. The first of these two gaps is partly closed here by distribution-free constructions that need only counts (Section III-G), in place of the resampling approach of Efron and Tibshirani [25]: exact intervals are now reported for every sensitivity and specificity, and Hanley-McNeil intervals for every AUC. What remains unavailable is anything that requires pairing: the absolute interval on a macro average, subgroup strata, calibration, and any model-to-model comparison. Table VI lists each estimand, its status, and the artifact change that would unblock it.

One consequence deserves emphasis. The difference between the reported macro AUC and any literature value is uninterpretable without an interval on the difference, and the ablation is not merely unquantified but unrun: no image-only checkpoint is registered, so no statement about the value of metadata fusion can be made from this study in either direction.

TABLE VI
Statistical estimands called for by the protocol, their status, the requirement that blocks each one, and what this paper reports instead.

Estimand	Status	Blocker	Reported instead
95% CI for sensitivity / specificity at 0.50	Reported	None: exact binomial inversion needs only the counts	Exact binomial, Table IV
95% CI for per-finding AUC	Reported	None: Hanley-McNeil variance needs only the AUC and class sizes	Hanley-McNeil, Table IV
95% CI for macro AUC (patient-level bootstrap)	Not computed	Per-study probabilities and patient IDs are discarded by the evaluation script; the variance of a mean of dependent AUCs has no closed form	Macro reported, no interval
Paired bootstrap: fused vs image-only	Not run	Requires an image-only checkpoint on the identical partition plus paired per-study predictions	No fusion claim (IV-F)
Subgroup metrics (sex, age band, view)	Not computed	Requires retained per-study metadata alongside predictions	Not reported
Calibration (reliability curve, ECE)	Not computed	Requires retained probabilities; ECE needs bin-wise counts	Not reported
Threshold selection on validation data	Not performed	No validation predictions were retained or used	Single fixed threshold, 0.50

Note: Intervals that need only counts are now reported; every quantity that needs pairing, resampling or per-study scores remains unavailable and is not estimated.

V. DISCUSSION AND LIMITATIONS

A. What the Results Support

The results support a narrow and defensible claim: on this subset and this split, a DenseNet121 plus metadata late-fusion model trained for eight epochs ranks held-out studies above chance for Pleural effusion (0.844) and Cardiomegaly (0.805), ranks them at a moderate level for Normal (0.724), and does so at a high-sensitivity, low-specificity operating point for effusion. They do not support a claim of clinical utility, which would require discrimination against a radiologist-annotated reference, a calibration curve, a decision-threshold analysis with explicit costs, and evidence of transportability to a second institution. They also do not support any statement about the contribution of the metadata channel, for the reason given in Section IV-F.

B. Scale, Covariate Shift and the Size of the Experiment

The experiment is small in four respects, and each bounds the interpretation. The cohort is 4,999 images drawn from a corpus of 112,120: less than five percent of the available data, and with no recorded selection rule. Four of fourteen findings are modeled, so the task is a projection of the benchmark rather than the benchmark. Training stopped after eight of twenty epochs, which by the training configuration means the model is under-trained rather than converged. And one seed was run, so the sensitivity of every number to initialization and data order is unknown. Finally the corpus is a single-institution collection: images were acquired on equipment and with protocols that do not represent the range of acquisition conditions in deployment, and a model validated only internally is expected to be optimistic relative to that deployment [13]. Covariate shift of this kind is the most likely explanation for the gap between the AUCs reported here and published ChestX-ray14 results for the same findings, which are typically obtained on the full corpus with a larger training budget; the gap is a statement about the experiment, not about the architecture.

C. Clinical Translation: Work, Not Assumption

Two things stand between this prototype and any clinical use, and neither is a modelling detail. The first is evidence cost: a reader study with radiologists, a calibration analysis, and external validation each require resources comparable to or larger than the training run itself, and the absence of any of them is a gap in the evidence rather than a gap in the code. The second is governance: a system that returns finding-level probabilities for chest radiographs would, in any deployment, be subject to regulatory review, prospective silent-mode evaluation and a defined route for acting on disagreements with the reporting radiologist. This paper does neither and claims neither.

D. Label Quality and the Threshold of Evidence

The error modes in Table V are more plausibly attributable to label construction than to model capacity. A study is labeled Normal when its report contains no positive finding, which conflates a genuinely normal examination with an under-reported one; a study is labeled Pneumonia only when the report uses that token, which is rare in a corpus where pneumonia is typically described descriptively. The asymmetric noise regime described by Karimi et al. [14] therefore applies: the positive class is comparatively clean, the negative class is contaminated, and the classifier is pushed toward over-prediction, which is the pattern of 212 Normal false negatives and 254 effusion false positives. This reading is an argument, not a measurement: no radiologist-annotated subset was used to estimate the noise rate. Under the argument, the largest available improvement is annotational rather than architectural, but the

paper does not claim to have demonstrated that.

E. The Fixed Threshold and the Missing Operating-Point Analysis

The fixed threshold of 0.50 is appropriate for an evaluation protocol, because it prevents the test set from being used to select an operating point, but it is not a clinical decision rule. With positive weighting the implicit prior differs across findings, so a single threshold places four decision boundaries at inconsistent positions on their ROC curves, as Table III shows. A deployment would require per-finding thresholds selected on validation data against explicit costs; the validation partition exists, was used for checkpoint selection, and was not used for this. Precision-recall curves, which summarize performance better than ROC at these prevalences, were not produced, and their absence is a gap that the retained artifacts cannot close without re-running evaluation.

F. Calibration, Subgroups and Explanations

No calibration analysis is reported. The weighted objective deliberately distorts predicted probabilities relative to empirical frequencies, so outputs must not be interpreted as risk [15]. On subgroup behaviour, the framework ingests age and sex as features and reports no stratified metrics; given the documented underdiagnosis bias of chest radiograph classifiers in under-served populations [16] and the gender-imbalance effects reported by Larrazabal et al. [17], the absence of a subgroup table is a substantive gap. Grad-CAM maps are produced by the service for qualitative inspection but are not shown here: the release contains neither radiographs nor a checkpoint, the framework includes no saliency sanity check, and post-hoc explanations should not carry the evidentiary weight of an interpretable model design [11]. A synthetic or illustrative saliency panel would be indistinguishable from fabricated evidence, so the paper omits the figure rather than approximating it.

G. Limitations

The following are stated as scope conditions, not as future work.

- **Single-corpus internal validation.** All reported numbers come from one institution's corpus and one patient-disjoint split. No external cohort, no cross-institution transport test and no temporal validation are reported, so the figures should be expected to be optimistic relative to deployment [13].
- **No bootstrap intervals and no paired comparison.** Patient-level bootstrap intervals and the paired ablation called for by the protocol could not be computed because per-study predictions were not retained; exact binomial and Hanley-McNeil intervals are reported instead, and they ignore within-patient clustering. Consequently no difference between this model and any other system, including its own ablation, is reported as statistically supported.
- **No control models.** No image-only model, no metadata-only model and no alternative backbone were trained. The study therefore contains no baseline, no ablation and no comparison with published results; the architecture's behaviour relative to any of these is unknown.
- **Small-event finding.** Pneumonia has six positive held-out studies. Its metrics are descriptive and unstable, its sensitivity interval spans 0.22–0.96, and the all-finding macro average is correspondingly dominated by that term.
- **Partially trained model.** Eight of twenty epochs completed with the stopping criterion unsatisfied. The run is under-trained and its numbers are an early-run result.
- **Single seed.** No seed replication, no hyperparameter search and no variance estimate across runs are reported.
- **Label noise of unknown magnitude.** Targets are report-derived; no radiologist-annotated subset was used to estimate the noise rate, and the direction of the resulting bias is argued rather than measured.
- **Subgroup performance not assessed.** No stratified analysis by sex, age band, view position or reported race/ethnicity is reported. Because the model consumes sex and age directly, subgroup performance cannot be assumed uniform [16]–[19].
- **Calibration not assessed.** No reliability diagram, expected calibration error or recalibration is reported, and the weighted objective biases the probabilities by construction [15].
- **Shortcut controls not run.** The metadata-only control, the view-position permutation audit and the adversarial-invariance variant described in Section IV-F were specified and not executed, so the metadata channel remains a plausible shortcut carrier rather than a measured one.
- **Design risks unquantified.** Horizontal flipping is applied to a laterally organized finding, checkpoint selection minimizes a weighted loss, and the training-partition prevalences that determine the weights are not retained, so none of these choices can be audited from the release.
- **Evaluation-record gaps.** Per-study predictions, ROC coordinates, training and validation loss series, patient counts per partition, training prevalences and the subset selection rule are all absent from the release. Section IV-G documents the one consequence that affects a reported number, and Table VI records the rest.
- **No reader study and no clinical validation.** No radiologist comparison, no time-to-decision measurement and no assessment of whether model output changes human performance are

available [4]. The system is a research prototype and must not be used for medical decision-making.

VI. CONCLUSION

This paper reported MediVision AI, a four-finding multimodal DenseNet121 baseline for chest radiograph classification, and an audit of the evaluation record that accompanies it. On a patient-disjoint 830-study held-out partition the framework reached a macro AUC of 0.771 over all four findings and 0.791 over the three findings with at least 30 positive studies, with the strongest discrimination for Pleural effusion (0.844) and Cardiomegaly (0.805) and the weakest, on six positive studies, for Pneumonia (0.709). At a fixed threshold of 0.50 the model operated at macro sensitivity 0.638 and macro specificity 0.705 over all findings, and at 0.628 and 0.767 over the three substantial findings. Exact intervals are reported for every sensitivity and specificity, so the precision of each estimate is visible rather than implied.

Four conclusions follow from the audit rather than from the score. First, the stored metric record is arithmetically closed and therefore usable: counts, rates and the macro average agree with one another exactly. Second, one archived ROC panel was not consistent with those counts, and the test that exposed it—an AUC that must lie inside the range attainable at its own operating point—is cheap enough that it should be part of any evaluation report. Third, the class-weighted objective cannot improve ranking metrics; it relocates the operating point instead, and the paper now states that instead of crediting the weighting with discrimination. Fourth, the claims that the current artifacts cannot support—a fusion benefit, a bootstrap interval, a calibrated probability, a subgroup difference—are all absent from this paper, and the artifact changes that would make each one computable are enumerated in Table VI.

The immediate work programme follows from that table and is short: persist per-study predictions and ROC coordinates at evaluation time; register image-only and metadata-only control checkpoints trained on the identical partition; complete the epoch budget with multiple seeds and select checkpoints by macro validation AUC; choose per-finding thresholds on the validation partition and report precision-recall curves; run the metadata permutation audit; and validate on a second institution's corpus. Until at least the first three are done, the framework is best described as a documented baseline for methodological study rather than as evidence about chest radiograph interpretation.

ACKNOWLEDGEMENT

The author thanks the maintainers of the public chest radiograph corpora used in this line of work for making de-identified research data available.

DECLARATIONS

Ethics. This study is a retrospective analysis of a public, de-identified secondary dataset (NIH ChestX-ray14, released under its own data-use agreement). It involved no prospective recruitment, no intervention and no access to identifiable patient information, so no new ethics approval or informed consent was required for the analysis reported here. Any clinical deployment would require ethics approval, regulatory review and prospective evaluation, none of which has been sought.

Data and code availability. The implementation is released as a Python/PyTorch modelling package together with the archived evaluation record and the figures of this paper. The image corpus is not redistributed; it is available from its original providers under their own agreements. Model checkpoints and the corpus are excluded from version control, which is the specific reason an independent re-execution of the evaluation is not possible from the release alone (Section IV-H). A short reproduction script regenerates every table and figure in this paper from the released evaluation record, and every number in the tables is emitted by it rather than typed by hand.

Clinical disclaimer. MediVision AI is a research prototype. Its outputs are model probabilities and not diagnoses, and Grad-CAM maps are attention diagnostics and not localizations of disease. The system must not be used for medical decision-making.

APPENDIX: CLAIM-TO-IMPLEMENTATION VERIFICATION

Each quantitative claim in Sections III and IV was checked directly against the implementation or the stored record; Table VII records the outcome. Two entries removed content rather than confirming it. The evaluation stage retains no per-study prediction (V8), which is why the bootstrap intervals and the ablation cannot be computed; and no control model was trained (V9) and one archived ROC panel does not match the counts (V11), which is why the paper makes no fusion claim and plots no archived curve. Every other entry confirms that the methods text now describes the code.

TABLE VII
Claim-to-implementation verification. Each row was checked directly against the source; rows are ordered by the section they support.

Item	Claim in the manuscript	Implementation or record as verified	Section
V1	Positive weighting is pos_weight = N_{neg}/N_{pos} per finding	Verified: per-finding counts are accumulated over the training loader and the ratio is passed to BCEWithLogitsLoss with denominators floored at 1 (Eq. 3)	III-E
V2	Augmentation is affine and horizontal flip on training inputs only	Verified: RandomAffine(degrees 5, translate 0.02, scale 0.97–1.03) and RandomHorizontalFlip(p 0.5); evaluation chain is resize and tensor conversion only	III-C
V3	Normalization uses ImageNet mean and SD in both chains	Verified: mean (0.485, 0.456, 0.406), SD (0.229, 0.224, 0.225)	III-C
V4	Splitting is patient-level, 70/15/15, seed 42	Verified by construction: the split is taken on unique patient identifiers; realized partition 3,447 / 722 / 830 images	III-B
V5	Evaluation uses one fixed threshold with no test-time tuning	Verified: 0.50 is a constant in the metric function and no threshold search touches the test partition	III-G
V6	Twenty epochs were requested; eight completed	Verified: the configuration sets 20 epochs and patience 5; the run record reports 8 completed epochs with the stopping criterion unsatisfied	III-F
V7	Checkpoints carry the per-epoch loss history and support resume	Verified in code (model, optimizer, scheduler, epoch index, validation loss, history); no checkpoint is part of the release, so no history can be shown	III-H
V8	Evaluation retains enough to support bootstrap intervals and an ablation	Contradicted by the record: the evaluation entry point writes aggregate metrics and count matrices only, discarding per-study probabilities and patient identifiers	IV-H
V9	Metadata fusion contributes to performance	Unevidenced: no image-only or metadata-only checkpoint is registered, so no comparison exists in either direction	IV-F
V10	Every reported metric is recomputable from the released artifacts	Verified: all twenty stored quantities reproduce from the stored count matrices to machine precision, and every interval in Table IV is recomputed from those counts by the routines of Section III-G	IV-G
V11	The archived ROC panel for Cardiomegaly describes the reported run	Contradicted: the archived curve reaches TPR 0.96 at FPR 0.137 where the counts place the operating point at TPR 0.479; an AUC of 0.957 lies outside the attainable range 0.447–0.928 of Eq. (7)	IV-G

Note: Items V8, V9 and V11 removed or qualified content rather than confirming it. Items V1 to V7 confirm that the methods text is an accurate description of the code. Table VI records the artifact changes that would let V8 and V9 be answered.

REFERENCES

[1] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, “ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases,” *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 3462–3471, 2017. doi: 10.1109/CVPR.2017.369.

Note: Crossref lists pp. 3462–3471 for the IEEE proceedings version; the widely circulated open-access PDF shows 2097–2106, which is the arXiv pagination and must not be cited.

[2] A. E. W. Johnson, T. J. Pollard, S. J. Berkowitz, N. R. Greenbaum, M. P. Lungren, C.-y. Deng, R. G. Mark, and S. Horng, “MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports,” *Sci. Data*, vol. 6, no. 1, pp. Art. no. 317, 2019. doi: 10.1038/s41597-019-0322-0.

[3] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya, J. Seekins, D. A. Mong, S. S. Halabi, J. K. Sandberg, R. Jones, D. B. Larson, C. P. Langlotz, B. N. Patel, M. P. Lungren, and A. Y. Ng, “CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison,” *Proc. AAAI Conf. Artif. Intell.*, vol. 33, no. 1, pp. 590–597, 2019. doi: 10.1609/aaai.v33i01.3301590.

Note: Crossref records this entry only under the AAAI proceedings DOI. The commonly quoted CVPR 2019 DOI (10.1109/CVPR.2019.00021) does not resolve in Crossref and was therefore not used.

[4] P. Rajpurkar, J. Irvin, R. L. Ball, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. P. Langlotz, B. N. Patel, K. W. Yeom, K. Shpanskaya, F. G. Blankenberg, J. Seekins, T. J. Amrhein, D. A. Mong, S. S. Halabi, E. J. Zucker, A. Y. Ng, and M. P. Lungren, “Deep learning for chest radiograph diagnosis: A retrospective comparison of the CheXNeXt algorithm to practicing radiologists,” *PLOS Med.*, vol. 15, no. 11, pp. Art. no. e1002686, 2018. doi: 10.1371/journal.pmed.1002686.

[5] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 2261–2269, 2017. doi: 10.1109/CVPR.2017.243.

Note: IEEE proceedings pagination is 2261–2269; the open-access version of record is paginated 4700–4708.

[6] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 770–778, 2016. doi: 10.1109/CVPR.2016.90.

[7] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, “ImageNet: A large-scale hierarchical image database,” *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.

[8] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, pp. 618–626, 2017. doi: 10.1109/ICCV.2017.74.

[9] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, “Learning deep features for discriminative localization,” *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 2921–2929, 2016. doi: 10.1109/CVPR.2016.319.

[10] L. Oakden-Rayner, “Exploring large-scale public medical image datasets,” *Acad. Radiol.*, vol. 27, no. 1, pp. 106–112, 2020. doi: 10.1016/j.acra.2019.10.006.

[11] C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” *Nature Mach. Intell.*, vol. 1, no. 5, pp. 206–215, 2019. doi: 10.1038/s42256-019-0048-x.

[12] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Mach. Intell.*, vol. 2, no. 11, pp. 665–673, 2020. doi: 10.1038/s42256-020-00257-z.

[13] J. R. Zech, M. A. Badgeley, M. Liu, A. B. Costa, J. J. Titano, and E. K. Oermann, “Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study,” *PLOS Med.*, vol. 15, no. 11, pp. Art. no. e1002683, 2018. doi: 10.1371/journal.pmed.1002683.

[14] D. Karimi, H. Dou, S. K. Warfield, and A. Gholipour, “Deep learning with noisy labels: Exploring techniques and remedies in medical image analysis,” *Med. Image Anal.*, vol. 65, pp. Art. no. 101759, 2020. doi: 10.1016/j.media.2020.101759.

[15] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, “Calibration: The Achilles heel of predictive analytics,” *BMC Med.*, vol. 17, no. 1, pp. Art. no. 230, 2019. doi: 10.1186/s12916-019-1466-7.

[16] L. Seyyed-Kalantari, H. Zhang, M. B. A. McDermott, I. Y. Chen, and M. Ghassemi, “Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations,” *Nature Med.*, vol. 27, no. 12, pp. 2176–2182, 2021. doi: 10.1038/s41591-021-01595-0.

[17] A. J. Larrazabal, N. Nieto, V. Peterson, D. H. Milone, and E. Ferrante, “Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis,” *Proc. Nat. Acad. Sci. USA*, vol. 117, no. 23, pp. 12592–12594, 2020. doi: 10.1073/pnas.1919012117.

[18] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, “Dissecting racial bias in an algorithm used to manage the health of populations,” *Science*, vol. 366, no. 6464, pp. 447–453, 2019. doi: 10.1126/science.aax2342.

[19] J. W. Gichoya, I. Banerjee, A. R. Bhimireddy, J. L. Burns, L. A. Celi, L.-C. Chen, R. Correa, N. Dullerud, M. Ghassemi, S.-C. Huang, P.-C. Kuo, M. P. Lungren, L. J. Palmer, B. J. Price, S. Purkayastha, A. T. Pyrros, L. Oakden-Rayner, C. Okechukwu, L. Seyyed-Kalantari, H. Trivedi, R. Wang, Z. Zaiman, and H. Zhang, “AI recognition of patient race in medical imaging: A modelling study,” *Lancet Digit. Health*, vol. 4, no. 6, pp. e406–e414, 2022. doi: 10.1016/S2589-7500(22)00063-2.

[20] T. Baltrusaitis, C. Ahuja, and L.-P. Morency, “Multimodal machine learning: A survey and taxonomy,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, pp. 423–443, 2019. doi: 10.1109/TPAMI.2018.2798607.

[21] S.-C. Huang, A. Pareek, S. Seyyedi, I. Banerjee, and M. P. Lungren, “Fusion of medical imaging and electronic health records using deep learning: A systematic review and implementation guidelines,” *npj Digit. Med.*, vol. 3, no. 1, pp. Art. no. 136, 2020. doi: 10.1038/s41746-020-00341-z.

[22] D. S. Kermany, M. Goldbaum, W. Cai, C. C. S. Valentim, H. Liang, S. L. Baxter, A. McKeown, G. Yang, X. Wu, F. Yan, J. Dong, M. K. Prasadha, J. Pei, M. Y. L. Ting, J. Zhu, C. Li, S. Hewett, J. Dong, I. Ziyar, A. Shi, R. Zhang, L. Zheng, R. Hou, W. Shi, X. Fu, Y. Duan, V. A. N. Huu, C. Wen, E. D. Zhang, C. L. Zhang, O. Li, X. Wang, M. A. Singer, X. Sun, J. Xu, A. Tafreshi, M. A. Lewis, H. Xia, and K. Zhang, “Identifying medical diagnoses and treatable diseases by image-based deep learning,” *Cell*, vol. 172, no. 5, pp. 1122–1131.e9, 2018. doi: 10.1016/j.cell.2018.02.010.

[23] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019. doi: 10.48550/arXiv.1711.05101.

[24] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, “PyTorch: An imperative style, high-performance deep learning library,” *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, pp. 8024–8035, 2019. doi: 10.48550/arXiv.1912.01703.

[25] B. Efron and R. J. Tibshirani, “An Introduction to the Bootstrap,” *New York, NY, USA: Chapman & Hall/CRC*, 1994. doi: 10.1201/9780429246593.